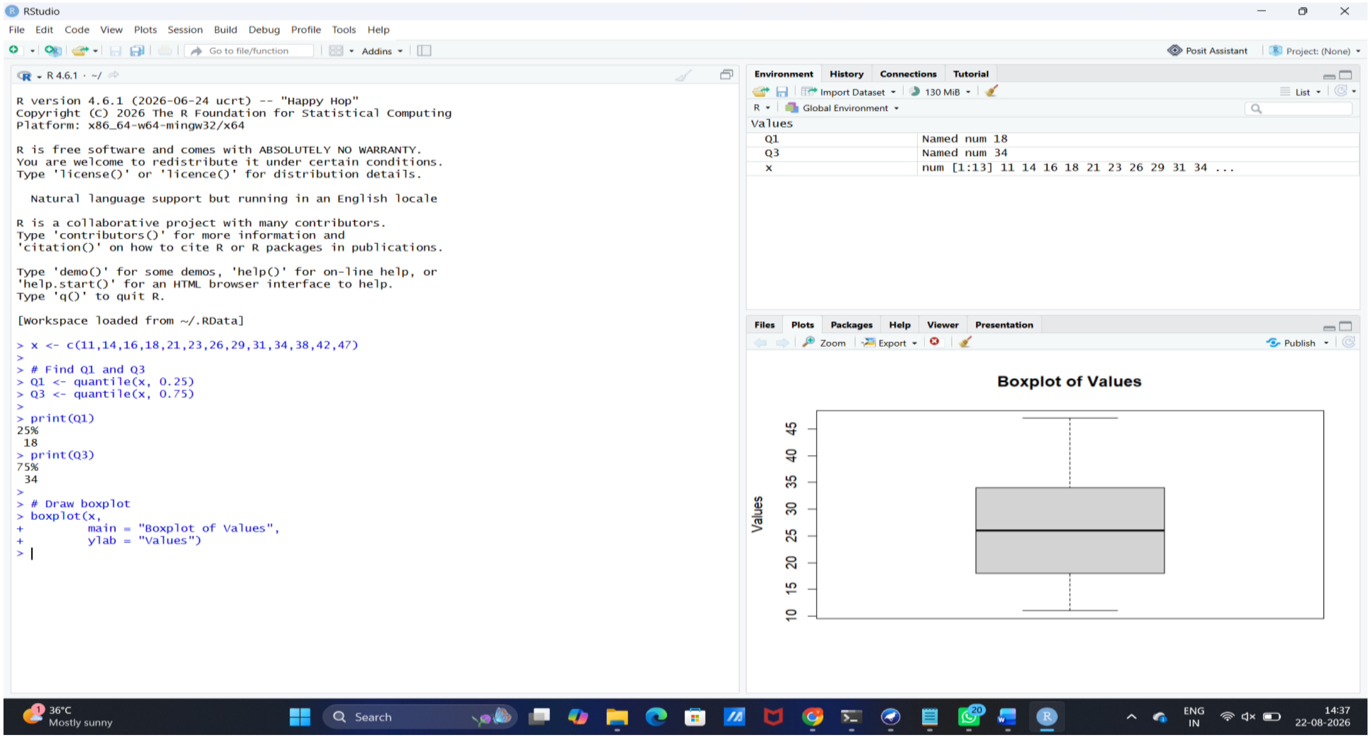
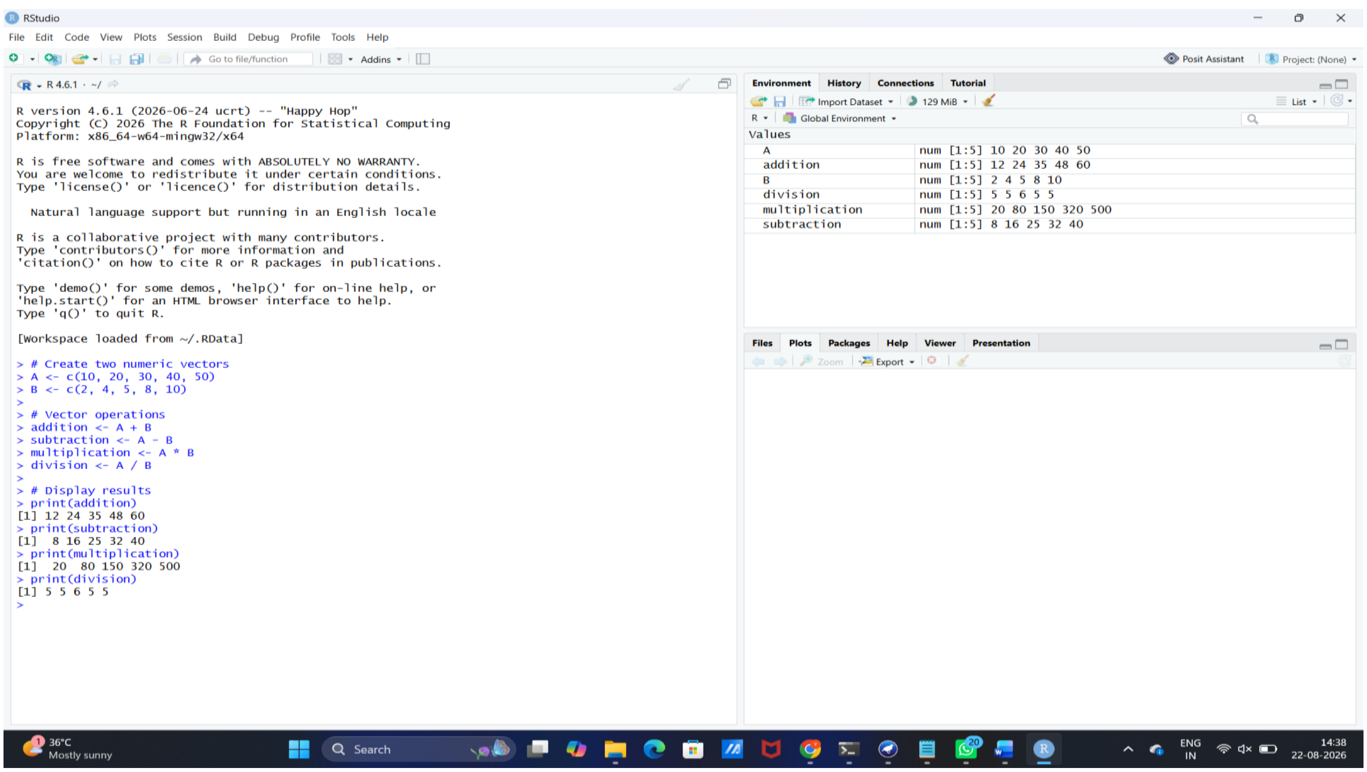
SET 16

1. R: Find Q1 and Q3 for 11,14,16,18,21,23,26,29,31,34,38,42,47 and draw a boxplot.



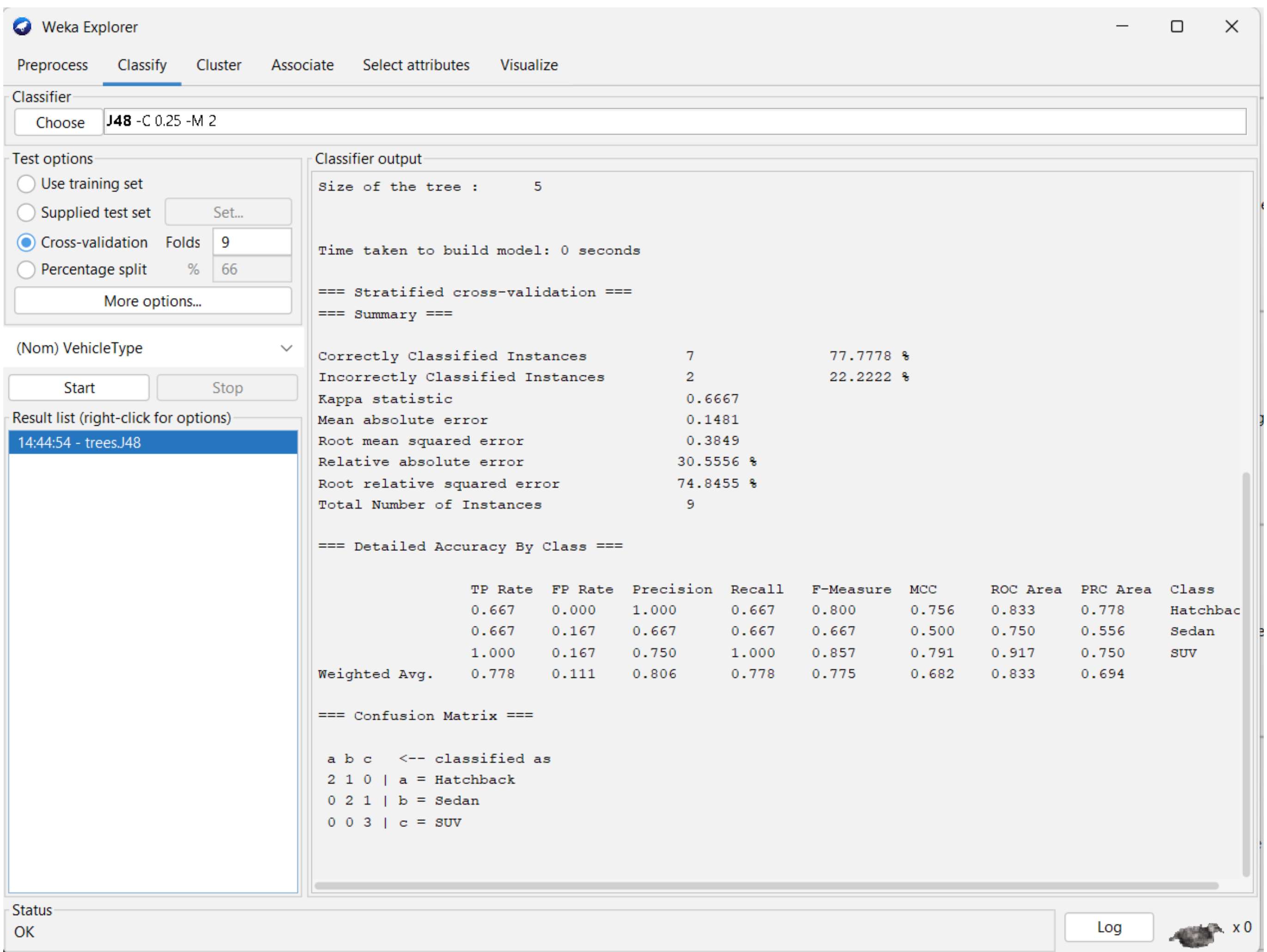
2. R: Demonstrate vector addition, subtraction, multiplication and division on two numeric vectors.



3. WEKA: Create vehicle-classification ARFF data (EnginePower, Weight, Mileage, VehicleType); apply J48 and inspect rules.

Dataset: Vehicle Classification

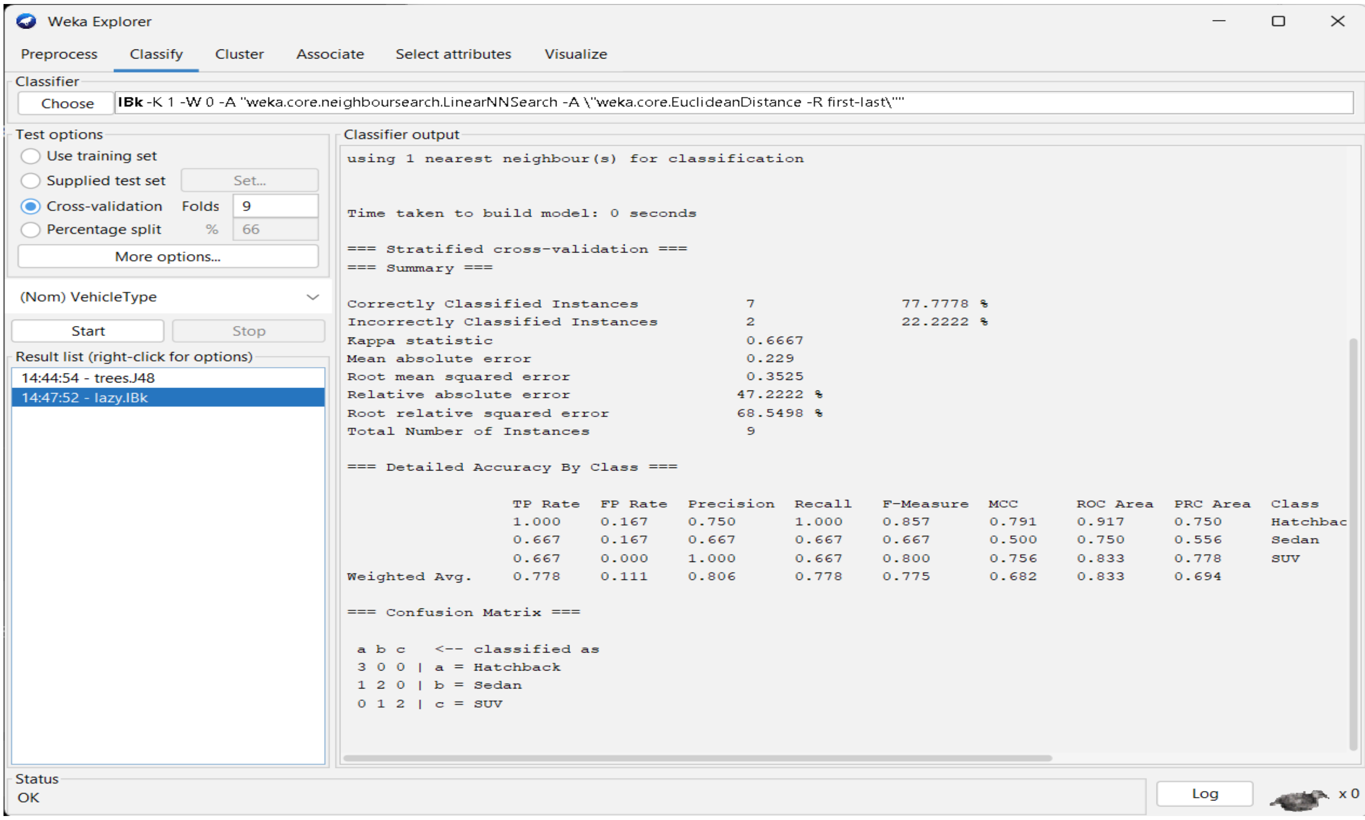
```
@relation vehicle
@attribute EnginePower numeric
@attribute Weight numeric
@attribute Mileage numeric
@attribute VehicleType {Hatchback,Sedan,SUV}
@data
70,900,20,Hatchback
90,1100,18,Sedan
120,1500,14,SUV
80,950,19,Hatchback
100,1250,16,Sedan
140,1800,12,SUV
75,920,21,Hatchback
110,1300,15,Sedan
150,1900,11,SUV
```

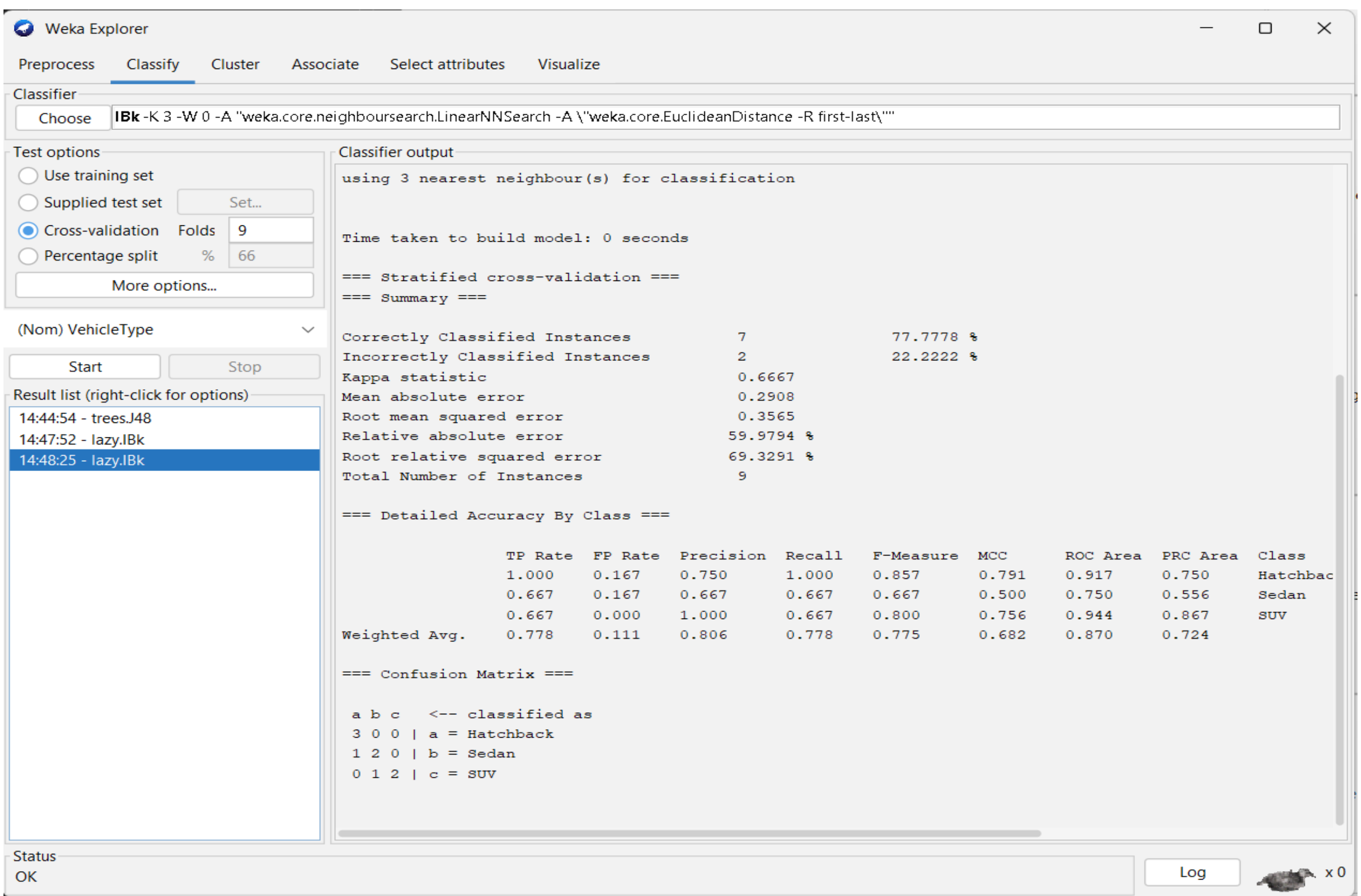


4. WEKA: Apply IBk to the vehicle data with two different K values and compare accuracy.

Dataset: Vehicle Classification – IBk

Use the same Vehicle Classification ARFF dataset from Set 16 Q3.


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Classifier
Choose
IBk -K 1 -W 0 -A "weka.core.neighboursearch.LinearNNSearch -A \"weka.core.EuclideanDistance -R first-last\""
Test options
Use training set
Supplied test set
Set...
Cross-validation Folds 9
Percentage split % 66
More options...
(Nom) VehicleType
Start
Stop
Result list (right-click for options)
14:44:54 - trees.J48
14:47:52 - lazy.IBk
Classifier output
using 1 nearest neighbour(s) for classification
Time taken to build model: 0 seconds
=== Stratified cross-validation ===
=== Summary ===
Correctly Classified Instances 7 77.7778 %
Incorrectly Classified Instances 2 22.2222 %
Kappa statistic 0.6667
Mean absolute error 0.229
Root mean squared error 0.3525
Relative absolute error 47.2222 %
Root relative squared error 68.5498 %
Total Number of Instances 9
=== Detailed Accuracy By Class ===
TP Rate FP Rate Precision Recall F-Measure MCC ROC Area PRC Area Class
1.000 0.167 0.750 1.000 0.857 0.791 0.917 0.750 Hatchbac
0.667 0.167 0.667 0.667 0.667 0.500 0.750 0.556 Sedan
0.667 0.000 1.000 0.667 0.800 0.756 0.833 0.778 SUV
Weighted Avg. 0.778 0.111 0.806 0.778 0.775 0.682 0.833 0.694
=== Confusion Matrix ===
a b c <-- classified as
3 0 0 | a = Hatchback
1 2 0 | b = Sedan
0 1 2 | c = SUV
Status
OK
Log
x 0


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Classifier
Choose
IBk -K 3 -W 0 -A "weka.core.neighboursearch.LinearNNSearch -A \"weka.core.EuclideanDistance -R first-last\""
Test options
Use training set
Supplied test set
Set...
Cross-validation Folds 9
Percentage split % 66
More options...
(Nom) VehicleType
Start
Stop
Result list (right-click for options)
14:44:54 - trees.J48
14:47:52 - lazy.IBk
14:48:25 - lazy.IBk
Classifier output
using 3 nearest neighbour(s) for classification
Time taken to build model: 0 seconds
=== Stratified cross-validation ===
=== Summary ===
Correctly Classified Instances 7 77.7778 %
Incorrectly Classified Instances 2 22.2222 %
Kappa statistic 0.6667
Mean absolute error 0.2908
Root mean squared error 0.3565
Relative absolute error 59.9794 %
Root relative squared error 69.3291 %
Total Number of Instances 9
=== Detailed Accuracy By Class ===
TP Rate FP Rate Precision Recall F-Measure MCC ROC Area PRC Area Class
1.000 0.167 0.750 1.000 0.857 0.791 0.917 0.750 Hatchbac
0.667 0.167 0.667 0.667 0.667 0.500 0.750 0.556 Sedan
0.667 0.000 1.000 0.667 0.800 0.756 0.944 0.867 SUV
Weighted Avg. 0.778 0.111 0.806 0.778 0.775 0.682 0.870 0.724
=== Confusion Matrix ===
a b c <-- classified as
3 0 0 | a = Hatchback
1 2 0 | b = Sedan
0 1 2 | c = SUV
Status
OK
Log
x 0